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Concise Report

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Copper/Photoredox-Catalyzed Regioselective 1,4-Addition to 1,3-Enynes: A Convenient Access to Perfluoroalkylated Allenes

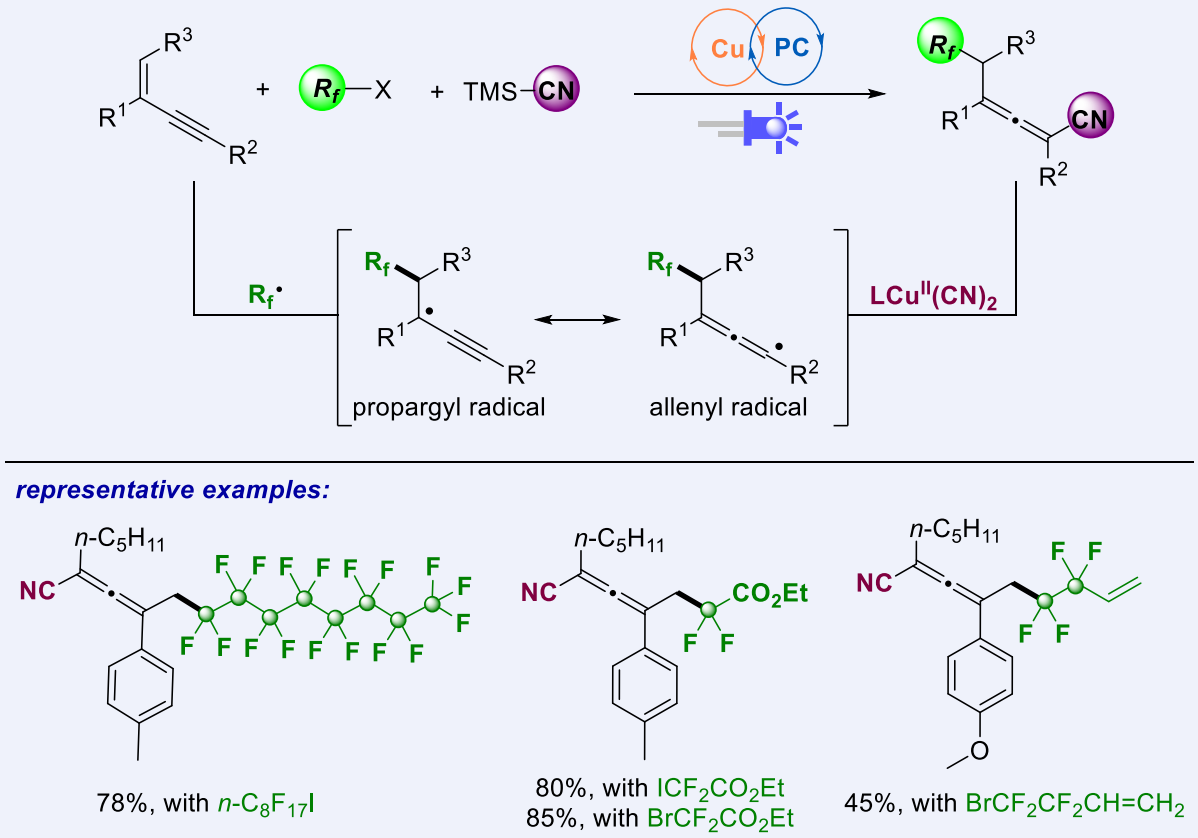
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Keywords

Fluorinated allenes | 1,3-Enynes | Photoredox | Fluoroalkylation | Multicomponent reactions | Difunctionalization | Radicals | Regioselectivity

Comprehensive Summary



A general and broadly applicable copper and photoredox dual-catalyzed multicomponent 1,4-perfluoroalkylcyanation of 1,3-enynes has been developed. This protocol enjoys success with high regioselectivity, mild reaction conditions, and excellent functional-group tolerance, allowing the facile synthesis of structurally diverse perfluoroalkylated allenes from readily available fluoroalkyl halides, 1,3-enynes and TMSCN in a one-pot manner. A reasonable mechanism has been proposed according to a series of control experiments.

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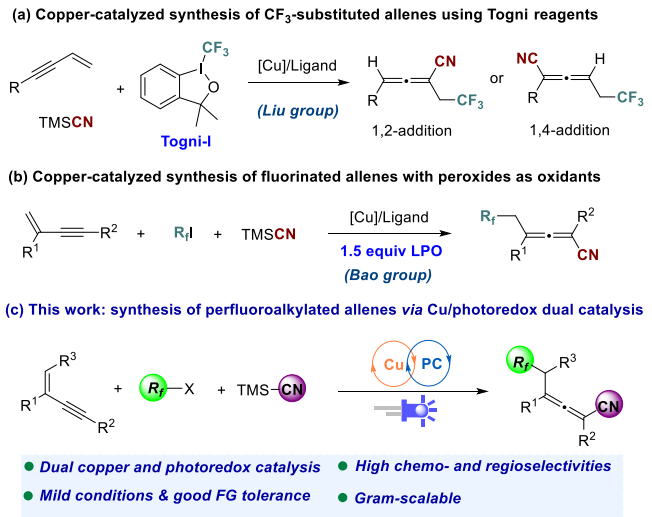
Background and Originality Content

The incorporation of a fluorine atom or fluorinated groups into organic frameworks can significantly alter their physicochemical properties and bioactivity, increasing their applicability as pharmaceuticals, agrochemicals, or building blocks for organic materials.[1] As a consequence, the development of efficient fluoroalkylation methods has attracted notable attention in drug discovery and modern organic chemistry.[2] In contrast to the recent significant progress in the fluoroalkylation of conventional alkenes and alkynes, which provides an access to the corresponding fluoroalkanes or fluoroalkenes,[3] efficient routes for the synthesis of fluoroalkylated allenes are less abundant. In recent years, di- and trisubstituted trifluoromethyl-allenes were successfully assembled by elimination,[4] isomerization,[5] rearrangements,[6] and S_N2' reactions[7] from alkynyl or alkenyl substrates bearing a CF_3 group. Alternative access to CF_3-allenes relied on the trifluoromethylation of propargylic derivatives with "$CuCF_3$" as a nucleophile.[8] Despite these important achievements, there are suffering limitations with the need for the prefunctionalized substrates, which are generally associated with low efficiency and poor atom economy. On the other hand, the installation of other fluorinated groups, such as perfluoroalkyl and functionalized difluoromethylene into allenes, particularly tetra-substituted allenes, are still remaining challenges.[9]

Multicomponent reactions (MCRs) represent an appealing strategy for the construction of multiple carbon-carbon or carbon-heteroatom bonds from easily available feedstocks in one simple operation.[10] Recently, the feasibility of MCRs in the assembly of multi-substituted allenes has been manifested through transition-metal-catalyzed difunctionalization of 1,3-enynes,[11] especially via radical-involved pathways.[12] In this regard, diverse kinds of difunctionalization, including alkyl/oxycyanation,[13] carbonsulfonylation,[14] alkylarylation,[15] trifluoromethylation[16] and others[17] have been achieved, offering a convenient access to functionalized allenes. Particularly, the group of Liu[18] realized the rapid construction of CF_3-containing allenyl nitriles using Togni reagents as the source of trifluoromethyl group (Scheme 1a). Besides, Bao and co-workers[19] reported the copper-catalyzed 1,4-fluoroalkylcyanation of 1,3-enynes with LPO as oxidative additives, delivering a series of fluorinated allenyl nitriles (Scheme 1b). The above-mentioned approaches either rely on the use of Togni reagents, or require stoichiometric external oxidants, which may decrease the compatibility of functional groups.

Given the high importance of fluorinated allenes and our continuous interest in difunctionalization of 1,3-enynes under visible-light-mediated photoredox conditions,[20] we sought to investigate the possibility of a one-pot 1,4-perfluoroalkylcyanation of 1,3-enynes in a redox neutral form. Herein, we disclose a novel Cu/photoredox catalyzed three-component reaction under mild conditions, in which two new carbon-carbon bonds are stimultaneously formed via 1,4-addition of perfluoroalkyl halides and TMSCN to 1,3-enynes, furnishing structurally diverse perfluoroalkylated allenes in a highly chemo- and regioselective manner (Scheme 1c).

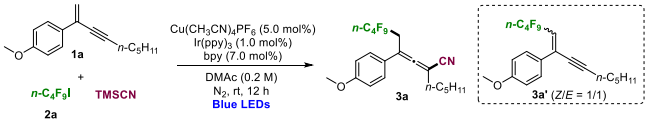
Scheme 1 Synthesis of fluorinated allenyl nitriles via radical difunctionalization of 1,3-enynes



Results and Discussion

The addition of *n*-perfluoroalkyl iodide **2a** and TMSCN to 1,3-enyne **1a** was selected as model reaction (Table 1). After a systematic investigation of reaction solvents, copper catalysts, ligands and photosensitizers (see the SI for details), the perfluoroalkylated allene **3a** was isolated in 73% yield when employing $Cu(CH_3CN)_4PF_6$ as the catalyst, bpy as the ligand and $Ir(ppy)_3$ as the photosensitizer in DMAc under blue LEDs irradiation at room temperature for 12 h (entry 1). It's worth noting that the addition exhibited high chemo- and regioselectivity, and no competing two-component cross-coupling byproducts, 1,2-addition or 3,4-addition byproducts were detected. When the amount of catalyst was reduced to 2.5 mol% $Cu(CH_3CN)_4PF_6$ and 3.5 mol% bpy, product **3a** was obtained with a slightly decreased yield (entry 2). As expected, control reactions omitting copper catalyst, ligand, photosensitizer or light indicated that each component was necessary for this cascade reaction (entries 3—6). A significant erosion in yield was observed when the reaction was conducted in air atmosphere (entry 7).

Table 1 Optimization of the reaction condition[a]



Entry	Variation from the standard conditions	Conv. of **1a**[b]/%	Yield of (**3a** / **3a'**)[c]/%
1	None	100	73 / trace
2	2.5 mol% [Cu], 3.5 mol% bpy	88	66 / trace
3	No $Cu(CH_3CN)_4PF_6$	46	0 / 35
4	No bpy	55	trace / 32
5	No $Ir(ppy)_3$	< 10	trace / 0
6	No light	< 10	0 / 0
7	In air	62	40 / trace

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.4 mmol), TMSCN (0.4 mmol), $Cu(CH_3CN)_4PF_6$ (5 mol%), bpy (7 mol%), $Ir(ppy)_3$ (1 mol%), and DMAc (1.0 mL), under N_2 at room temperature, 35 W blue LEDs, 12 h. [b] Determined by GC analysis using dodecane as internal standard. [c] Isolated yield.

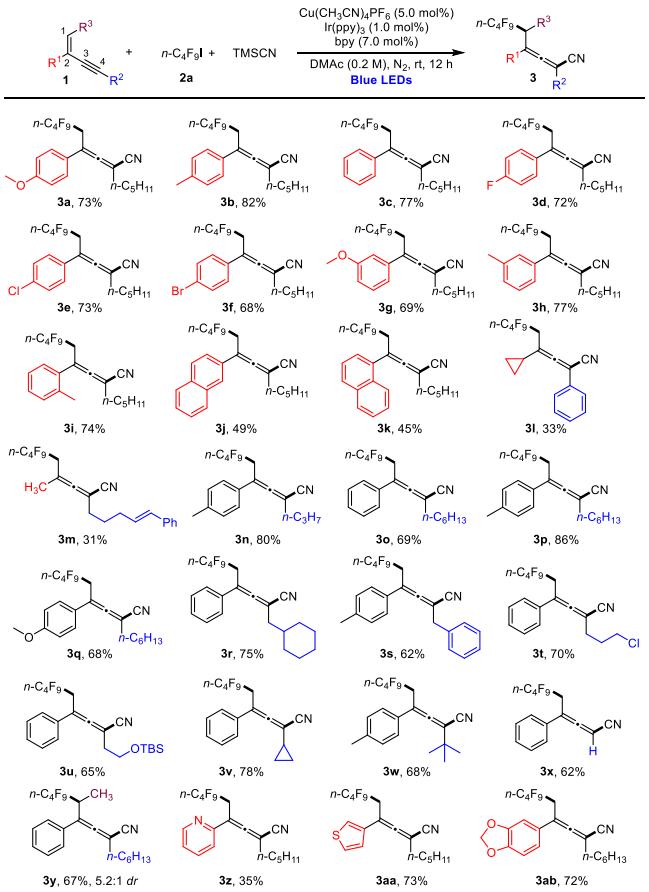
With the optimal reaction parameters in hand, we explored the substrate scope of 1,3-enynes (Scheme 2). The assessment started with electronic effects on the C2-aromatic ring. The 1,3-enynes bearing electron-donating (-OMe, -Me), electron-neutral (-H) and electron-withdrawing substituents (-F, -Cl, -Br) on the *para*-position of benzene ring furnished the corresponding products **3a**—**3f** with 68%—82% yields. The *meta*-position with methoxy or methyl group could afford satisfactory yields (**3g** and **3h**). To our delight, the *ortho*-position with methyl group also delivered the product **3i** in 74% yield. Sterically demanding substrates could work smoothly as well, providing the products **3j** and **3k** in modest yields. With respect to the C2-alkyl substituents, 1,3-enynes embodying a cyclopropyl or methyl group were successfully employed for this reaction, albeit with lower yields (**3l** and **3m**). The alkynyl moiety in the 1,3-enynes was also investi-

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gated. Linear alkyl chains with different lengths provided products **3n**—**3q** in 68%—86% yields. The branched alkyl group and benzyl substituent were proven to be compatible as well, affording the products **3r** and **3s** in good yields. In addition, the system tolerated various functional groups, including chlorine, OTBS, cyclopropyl and a remote alkene function, giving the allene products (**3t**—**3v** and **3m**) in acceptable yields. The steric property of the alkynyl moiety had no obvious effects on the transformation efficiency (**3w**). Particularly, monosubstituted 1,3-enyne at the 4-position was a good candidate as well, affording the trisubstituted allene **3x** in 62% yield. Furthermore, when 1,3-enyne possessing an internal alkene was employed, product **3y** was isolated in 67% yield with modest diastereoisomeric ratio (5.2 : 1). Moreover, 1,3-enynes bearing heterocycles such as pyridine, thiophene and 1,3-benzodioxole were also successfully employed for the reaction, and the corresponding products were achieved in positive yields (**3z**—**3ab**).

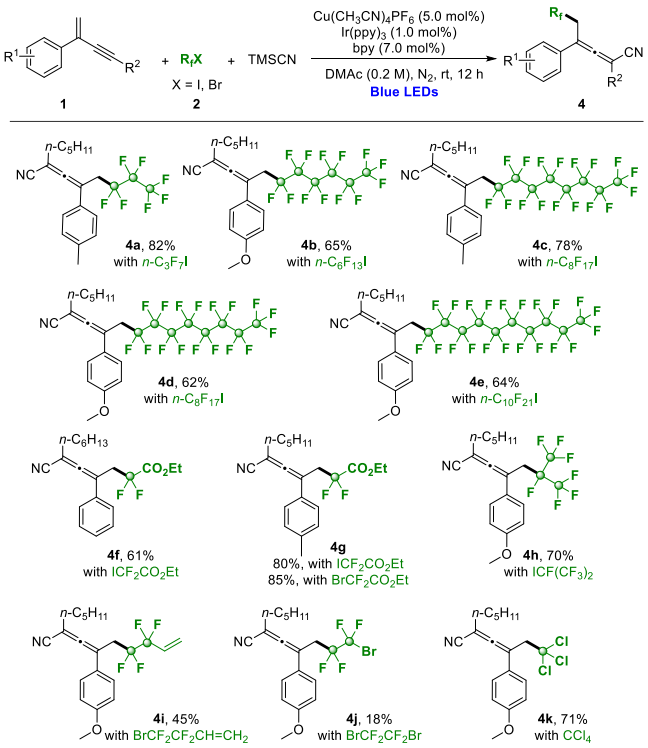
Scheme 2 Substrate scope with 1,3-enynes[a]



[a] Standard conditions: **1** (0.2 mmol), **2a** (0.4 mmol), TMSCN (0.4 mmol), $Cu(CH_3CN)_4PF_6$ (5 mol%), bpy (7 mol%), $Ir(ppy)_3$ (1 mol%), and DMAc (1.0 mL), under N_2 at room temperature, 35 W blue LEDs, 12 h. Isolated yield.

Then, we switched our attention to the scope of perfluoroalkyl halides. As shown in Scheme 3, perfluoroalkyl iodides with different lengths, from perfluoropropyl to perfluorodecyl, were all well tolerated in the 1,4-perfluoroalkylcyanation of 1,3-enynes, providing the corresponding allenes **4a**—**4e** in 62%—82% yields. Ethyl iododifluoroacetate could also participate in this transformation, affording difluoromethylene substituted allenes **4f** and **4g** in good yields. Notably, a comparable yield was obtained when using ethyl bromodifluoroacetate, a low-cost commercially available raw material as the coupling partner. The relatively sterically hindered perfluoroisopropyl group could be installed as well, and **4h** was achieved in 70% yield. Additionally, the fluoroalkylating reagent bearing a terminal alkene group was well tolerated, albeit with a modest yield (**4i**), providing an additional handle for further functionalization. When 1,2-dibromotetrafluoroethane was applied, a decreased yield was obtained (**4j**) and no diallenylation product was observed. To our surprise, tetrachloromethane was also a good candidate, furnishing trichloroallene product **4k** in useful efficiency. It is worth noting that no allene products **4i** and **4k** were obtained when the reaction was performed under the existing copper/peroxide system,[19] which preliminarily demonstrated the advantage of the mild copper/photoredox dual catalysis.

Scheme 3 Substrate scope with fluoroalkylated halides[a]



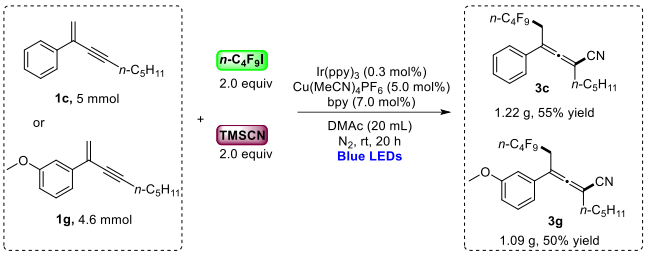
[a] Standard conditions: **1** (0.2 mmol), **2** (0.4 mmol), TMSCN (0.4 mmol), $Cu(CH_3CN)_4PF_6$ (5 mol%), bpy (7 mol%), $Ir(ppy)_3$ (1 mol%), and DMAc (1.0 mL), under N_2 at room temperature, 35 W blue LEDs, 12 h. Isolated yield.

To further demonstrated the practical utility of this transformation, gram-scale reactions were carried out under mild conditions, affording the perfluoroalkylated allenes **3c** and **3g** in 55% and 50% isolated yields, respectively, with a small loss of efficiency (Scheme 4). Subsequently, we conducted several control experiments to gain more insight into the 1,4-perfluoroalkylcyanation pathway. As shown in Scheme 5a, the reaction was completely inhibited when the radical scavenger TEMPO (2,2,6,6-tetramethyl-1-piperidinyloxy) or 1,1-diphenylethylene was added under the standard conditions, and the corresponding radical-trapping products **5a** and **5b** were obviously detected by GC-MS, indicating the involvement of the fluoroalkyl radical. In addition, ring-opened product **5c** was obtained in a similar yield when 1,3-enyne **1z** and *n*-perfluoroalkyl iodide **2a** were subjected to the reaction in the presence or absence of the copper catalyst (Scheme 5b). This observation supports that the fluoroalkyl radical was generated through photoreduction. More importantly, these results demonstrate the existence of the propargylic radical intermediate, which was formed by fluoroalkyl radical addition and could deliver the allenyl radical via resonance. Finally, Stern–Volmer luminescence studies indicated that perfluoroalkyl iodides could quench the excited-state $*Ir^{III}$ efficiently, which demonstrated an oxidative quenching mechanism (see the SI for details).

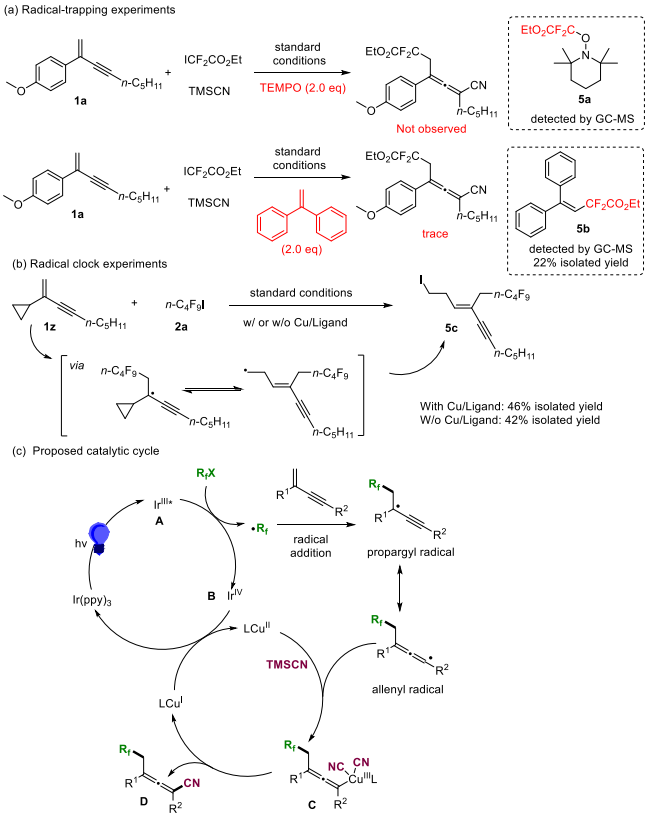
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Scheme 4 Gram-scale reactions



Scheme 5 Control experiments and proposed catalytic cycle



According to the above studies and related literature precedents,[18,20a] a plausible mechanism for photoredox and copper dual catalysis is suggested in Scheme 5c. Upon irradiation of blue LEDs, the excitation of photocatalyst $Ir(ppy)_3$ leads to the active state **A** $[Ir(ppy)_3]^*$ ($E_{1/2}^{III*/IV}$ = −1.73 V vs SCE), which then undergoes single electron transfer (SET) with perfluoroalkyl iodides (−1.0 to −1.5 V vs. SCE)[9] to afford fluoroalkyl radicals. The subsequent addition of the radical to the double bond of 1,3-enyne generates the propargyl radical, which is in resonance with the allenyl radical. Meanwhile, Cu(I) species is oxidized by high-valence photosensitizer **B** to afford Cu(II) species, which could react with TMSCN by ligand exchange. Then, the allenyl radical is captured to produce the allenyl copper(III) species **C**. Subsequent reductive elimination furnishes the perfluoroalkylated allene **D** and regenerates Cu(I) species to complete the catalytic cycle.

Conclusions

In conclusion, we have developed a visible-light-induced 1,4-perfluoroalkylcyanation of 1,3-enynes via copper and photoredox dual catalysis, which points the way to streamline the synthesis of functional perfluoroalkylated allenes. The reaction obviates the need for stoichiometric external oxidants and features mild conditions, broad functional-group tolerance, and exceptional chemo- and regioselectivities, allowing for facile access to gram-scale reactions. Preliminary mechanistic studies indicate that this transformation proceeds through a radical relay involving the resonance of the propargyl radical with the allenyl radical. Further studies of the detailed mechanism and additional applications of this protocol to synthesize fluorinated druglike molecules are ongoing.

Experimental

General procedure for the 1,4-perfluoroalkylcyanation of 1,3-enynes

In a 10 mL sealed tube, $Cu(CH_3CN)_4PF_6$ (3.7 mg, 0.01 mmol, 5.0 mol%), bpy (2.2 mg, 0.014 mmol, 7.0 mol%), and $Ir(ppy)_3$ (1.3 mg, 0.002 mmol, 1.0 mol%) were added in degassed DMAc under a nitrogen atmosphere, and the mixture was stirred at room temperature for 30 min. Then 1,3-enyne (0.2 mmol, 1.0 equiv), R_fX (0.4 mmol, 2.0 equiv) and TMSCN (50 μL, 0.4 mmol, 2.0 equiv) were sequentially added. The reaction mixture was stirred for 12 h in the presence of a 35 W Blue LED lamp (with a cooling fan to keep the reaction temperature at room temperature). After the reaction completion, the reaction mixture was diluted with EtOAc, then washed with brine, dried over $MgSO_4$, filtered, and concentrated under reduced pressure. The residue was purified by flash chromatography on silica gel to afford the allene products.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202400207.

Acknowledgement

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